# Home Assignment - 01
## Computer Networks

① During one transmission, Shraddha wants to transmit the 4-bit data word 1011 using a appropriate Hamming (7,4) code. Determine the number of parity bits required and justify your answer.

② Identify the position of data bits and parity bits in Hamming code..

Ans: Hamming (7,4) Code

Number of parity bits required.

For Hamming code: $2^P \geq m + P + 1$

Here, data bits $m = 4$

Try $P = 3$:

$$2^3 = 8 \geq 4 + 3 + 1 = 8$$

$\therefore$ Parity bits required $= 3$

② Positions of data bits and parity bits in Hamming (7,4):

| Position | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Type | P1 | P2 | D1 | P4 | D2. |

Data = 1011

So, $D1 = 1$, $D2 = 0$, $D3 = 1$, $D4 = 1$

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(b) Generate the complete 7-bit Hamming code for the given data word using even parity.

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Bit | P1 | P2 | 1 | P4 | 0 | 1 | 1 |

- P1 Checks (1, 3, 5, 7) → (P1, 1, 0, 1) → even → P1 = 0
- P2 Checks (2, 3, 6, 7) → (P2, 1, 1, 1) → odd → P2 = 1
- P4 Checks (4, 5, 6, 7) → (P4, 0, 1, 1) → even → P4 = 0

∴ Hamming code = 0 1 1 0 0 1 1

(c) Assume that during transmission, the received code word is 1010011.

- Detect whether an error has occurred.
- If an error is present, identify the position of the erroneous bit.

Received code word = 1010011

Check parity:

- P1 Checks → error
- P2 Checks → error
- P4 Checks → correct

error position = (P4 P2 P1) = $011_2$ = 3

→ error at position 3.

(d) Correct the error and retrieve the original 4-bit data word.

→ flip bit at position 3

Corrected Code = 1110011

exact data bits (positions 3, 5, 6, 7)

Original data = 1011.

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(2) A data communication systems uses a checksum technique for error detection. The data to be transmitted is divided into 4 blocks, each consisting of 8 bits, as shown below:

Block 1: 11011110 Block 2: 10110101 Block 3: 01101100
Block 4: 10011001 Compute the checksum.

<u>Block 1</u> : 11011110
10110101
01101100
10011001

<u>Binary addition</u>

```
  11011110
+ 10110101
----------
  100010011   (carry wrapped)

+ 01101100
----------
  11110111

+ 10011001
----------
  110010000   (carry wrapped)
```

- final sum : 10010001

Checksum : 1's complement

Checksum = 01101110

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③ The data bits to be transmitted are 111011, and the generator polynomial used is $G(x) = x^3+1$. Explain how the receiver uses the same generator polynomial to detect errors in the received frame.

Data = 111011

Generator polynomial = $G(x) = x^3+1 \rightarrow 1001$

Sender:

- Append 3 zeros → 111011000
- Divide by 1001
- Append remainder → transmitted frame

Receiver:

- Divide received frame by 1001
- If remainder = 000 → no error
- If remainder ≠ 000 → error detected

→ Receiver uses the same generator polynomial to verify data integrity.

④ A: 0100011 B: 1110011;
FLAG: 01111110, ESC: 11100000
Show the bit sequence transmitted (in binary) for the four-character frame.
"A FLAG B ESC" When each

Given:

A = 0100011
B = 1110011
FLAG = 01111110
ESC = 11100000

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Frame : A FLAG B ESC

(a) Byte count-framing
(Byte count = 4 → 00000100)

00000100 01000111 01111110 11100011
11100000

(b) flag Bytes with byte stuffing

01111110
01000111
11100000 01111110
11100011
11100000 11100000
01111110

→ flag at start and end, ESC before FLAG and ESC in data.

(c) flag bytes with Bit stuffing
(Insert 0 after five consecutive 1s)

01111110
01000111
011111010
11100011
11100000
01111110

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